

DEVSECOPS & PLATFORM · ISSUE 03 OF 08

GitOps with Flux and Argo CD.

Git is the change record. The cluster is a cache of it.

03

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Swipe →

The reconcile loop

- 1 Commit
- 2 Controller detects
- 3 Render Helm / Kustomize
- 4 Diff vs cluster
- 5 Apply
- 6 Health check

The loop runs continuously, not just on commit. Manual changes are reverted.

Repo structure

01 apps/base

Shared manifests for each service.

02 apps/overlays/<env>

Small per-environment differences.

03 clusters/<name>

Entry point each cluster reconciles.

04 infrastructure/

Controllers and CRDs, applied first.



Sync strategy by environment

ENV	STRATEGY
Dev	Auto-sync + self-heal + prune
Staging	Auto-sync, reviewed prune
Prod	PR approval is the gate
All	Dependencies: CRDs before apps



Flux vs Argo CD

FLUX

- CRD-native, CLI first
- Kustomization + HelmRelease
- No UI by default
- Image automation built in

ARGO CD

- Strong UI
- Applications + ApplicationSets
- SSO and RBAC for the UI
- Sync waves and hooks

GitOps habits

DO

- ✓ Promote with PRs between env folders
- ✓ Keep secrets encrypted or external
- ✓ Alert on sync failures

AVOID

- ✗ kubectl edit in production
- ✗ Plaintext secrets in Git
- ✗ One giant repo-wide sync

KEY TAKEAWAY

Promote with pull requests, recover with git revert.

- Continuous reconcile, not one-off apply
- Base + overlays + cluster entry points
- Prod gate is the PR review

YOUR TURN

Flux or Argo CD — and what made you choose?

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Next in the series: #04 Helm vs Kustomize

Save · Repost

